

### 3.2.1. Statistical Parity

Statistical parity requires that the probability of a positive outcome—in this context, the likelihood of a resume being selected as better—be equal across groups defined by a focal attribute. While this criterion is traditionally applied to protected human attributes such as gender or race, we adapt it here to study disparities across resumes defined by their source of generation: whether a resume was generated by the evaluator LLM or by an alternative source (a human or another LLM). In this setting, statistical parity captures unconditional differences in selection rates and serves as a descriptive measure of whether the evaluator LLM disproportionately selects its own outputs.

To operationalize this, we conduct pairwise resume comparisons involving resumes generated by either a human or one of several LLMs. Specifically, we test whether an LLM  $f$ , when serving as the evaluator LLM, is more likely to select a resume it generated over one written by a human, or over one produced by an alternative LLM, when the resumes are otherwise equivalent in content quality.

Let  $S \in \{0, 1\}$  denote the source indicator, where  $S = 1$  if the resume was generated by the evaluator LLM  $f$ , and  $S = 0$  if it was written by a human or generated by an alternative LLM. Let  $Y'_f \in \{0, 1\}$  denote the binary decision made by evaluator  $f$ , where  $Y'_f = 1$  indicates that the resume is selected as stronger. We define the Statistical Parity Self-Preference Bias as the difference in *selection rates* between resumes generated by the evaluator LLM and those from other sources:

$$\text{Statistical Parity Self-Preference Bias}_f = \mathbb{P}(Y'_f = 1 \mid S = 1) - \mathbb{P}(Y'_f = 1 \mid S = 0). \quad (1)$$

This formulation Eq. (1) captures the difference in the evaluator LLM  $f$ 's likelihood of selecting a resume it generated versus one it did not, providing a direct measure of unconditional AI self-preferencing. Interpreting this difference as bias, however, requires caution because it can reflect not only self-preference but also systematic differences in *content quality*. In this context, content quality refers to the effectiveness with which a resume communicates the same underlying candidate information—such as clarity, coherence, fluency, and organization—rather than differences in the candidate's qualifications or background. As a result, a positive statistical-parity gap may arise even in the absence of intrinsic self-preference if the evaluator LLM tends to produce higher-quality renderings of the same candidate information (for instance, when users rely on a more capable model to polish resume wording and structure).

### 3.2.2. Equal Opportunity

To address the limitation of the statistical parity measure and disentangle intrinsic self-preferencing from differences in content quality, we adopt the equal opportunity fairness criterion

of Hardt et al. (2016). The key idea underlying equal opportunity is to compare selection behavior conditional on merit. In our setting, this allows us to assess whether an evaluator LLM exhibits systematic self-preference even when resumes are of comparable content quality.

Let  $Y \in \{0, 1\}$  represent the ground truth quality label, where  $Y = 1$  indicates that the resume is of higher content quality and would be selected by a human evaluator. We define the Equal Opportunity Self-Preference Bias for evaluator LLM  $f$  as:

$$\text{Equal Opportunity Self-Preference Bias}_f = \mathbb{P}(Y'_f = 1 \mid S = 1, Y = 1) - \mathbb{P}(Y'_f = 1 \mid S = 0, Y = 1). \quad (2)$$

By conditioning on resume content quality, this formulation Eq. (2) isolates the evaluator LLM  $f$ 's intrinsic tendency to favor its own generated content. A positive value of this measure indicates that, even among resumes that are comparably high quality, the evaluator LLM  $f$  is more likely to select resumes it generated itself, providing evidence of self-preferencing that cannot be explained by superior content quality alone.

We operationalize the equal opportunity criterion using two complementary approaches that reflect the fact that content quality is a latent variable and cannot be directly observed. First, we use a regression model to control for observable linguistic and textual features, which serve as proxies for content quality in pairwise comparisons. Second, we rely on human annotators to independently evaluate each resume pair and determine which resume is better, using these judgments as the ground truth for the latent content quality.

**Conditional Logistic Regression.** In line with the rationale behind the equal opportunity, we estimate a conditional logistic regression model that conditions on each resume pair  $\mathcal{R}_i$ , thereby absorbing all pair-specific quality differences. Let  $i$  index resume pairs and  $j \in \{1, 2\}$  index the two resumes within each pair. The model is given by

$$\log \left( \frac{\mathbb{P}(\text{Preferred}_{ij} = 1 \mid \mathcal{R}_i)}{1 - \mathbb{P}(\text{Preferred}_{ij} = 1 \mid \mathcal{R}_i)} \right) = \beta_1 \cdot \text{evaluatorLLM}_{ij} + \underbrace{\beta_2^\top \phi_{ij} + \beta_3^\top \psi_{ij}}_{\text{Content Quality Controls}}, \quad (3)$$

where  $\text{Preferred}_{ij} = 1$  indicates that resume  $j$  is selected as better within pair  $i$ , and  $\text{evaluatorLLM}_{ij}$  is an indicator for whether resume  $j$  was generated by the evaluator LLM. Standard errors are clustered at the resume-pair level to account for within-pair correlation in residuals.

We include two sets of control variables that proxy for the latent quality of resume content. The first set, denoted by  $\phi_{ij}$ , comprises a rich collection of linguistic features drawn from the Linguistic Inquiry and Word Count (LIWC) framework (Boyd et al. 2022). These features capture multiple dimensions of writing style and psychological tone that are plausibly related to how effectively a resume conveys information. Specifically, we include measures of (i) length and lexical complexity, such as total word count and the prevalence of longer words; (ii) psychological tone and

professionalism, captured by Analytic, Clout, Authentic, and overall emotional Tone; (iii) linguistic structure, reflected in the use of function words and core grammatical categories (e.g., prepositions, auxiliary verbs, and conjunctions); and (iv) agency and action orientation, proxied by the use of personal pronouns (e.g., I, we, you, they) and the balance between verbs and adjectives.

The second set of controls, denoted by  $\psi_{ij}$ , captures semantic similarity and content preservation between the resume summary and the rest of the resume. We include BERTScore (Zhang et al. 2020), which measures semantic similarity using contextual embeddings from BERT, and ROUGE-L (Lin 2004), which captures sequence-level overlap based on longest common subsequences. These metrics help ensure that estimated effects are not driven by differences in informational content or semantic drift, but rather reflect differences in presentation and stylistic expression.

In the specification given in Eq. (3), the key parameter of interest is  $\beta_1$ , which captures the evaluator LLM’s tendency to prefer its own output, after controlling for observable content quality. A significantly positive estimate of  $\beta_1$  indicates that the evaluator LLM is more likely to select its own output over competing content, even after adjusting for measurable aspects of content quality. With this parameter, the equal opportunity self-preference bias can be computed with  $\frac{e^{\beta_1}}{1+e^{\beta_1}} - \frac{1}{1+e^{\beta_1}}$ , which represents the difference in the probability that an LLM evaluator prefers its own output over a competing alternative, holding content quality constant.

**Human Annotations.** Since measuring equal-opportunity self-preference bias requires ground-truth labels identifying which resume in a pair is of higher quality, we obtain human evaluations.

To this end, we recruit 18 human annotators from Prolific to evaluate two types of resume pairs: (1) human-written resumes versus their LLM-generated counterfactuals, and (2) resumes generated by one LLM versus another LLM. For both tasks, we focus on three representative LLMs: GPT-4o, DeepSeek-V3, and LLaMA-3-70B. Specifically, for case (1), the evaluation includes (i) GPT-4o vs. Human, (ii) DeepSeek-V3 vs. Human, and (iii) LLaMA-3-70B vs. Human. For case (2), we evaluate (i) DeepSeek-V3 vs. GPT-4o, (ii) DeepSeek-V3 vs. LLaMA-3-70B, and (iii) GPT-4o vs. LLaMA-3-70B. Each comparison condition is evaluated by three annotators.

Each annotator is presented with 30 resume pairs from one of the comparison cases above.<sup>2</sup> The number of evaluations per annotator is chosen to balance cognitive workload with fair compensation. To prevent confirmation bias, the order of the resumes within each pair is randomized, and annotators are blinded to the source of each resume. For each resume pair, annotators rate both resumes on five linguistic dimensions—clarity, fluency, coherence, conciseness, and overall quality—and are then asked to select which resume is better.

<sup>2</sup> In the annotation instructions, annotators were informed that there are 32 pairs in total, including 2 attention-check pairs that are excluded from the final analysis.